

John Leiga

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U.S. Citizen • Eligible to work on export-controlled programs

Education

B.S. Mechanical Engineering

2017 – 2021

University of California, Merced

Skills

Design and Analysis: Precision mechanical design, GD&T, ASME Y14.5, tolerance stackups, stress analysis, finite element analysis (FEA), design for manufacturing (DFM), design for assembly (DFA)

Manufacturing and Systems: Mechanical controls integration, root cause analysis, 5 Whys, Lean, Kaizen, process optimization, factory startup, production equipment validation, CapEx execution, contractor management

Testing and Validation: Environmental testing including vibration, shock, and thermal, instrumentation, data acquisition, failure analysis, design of experiment (DOE), capability analysis

Tools: SolidWorks, Autodesk Inventor, AutoCAD, Python, MATLAB, Minitab

Experience

Nestle Purina PetCare Company

Bloomfield, MO — Ft. Dodge, IA — St. Louis, MO

Design Engineer

Feb 2025 – Present

- Improved production systems across multiple facilities through process optimization and equipment troubleshooting
- Validated process equipment for factory startup, supporting installation readiness, startup checks, and production ramp
- Translated production needs into mechanical specifications, equipment requirements, tooling improvements, and operating standards
- Designed extrusion dies, production tooling, and equipment components in Autodesk Inventor for high-volume manufacturing
- Resolved production critical failures through root cause analysis, hands on troubleshooting, and standard work updates

Management Development Associate - Engineering

May 2023 – Feb 2025

- Stabilized a 14-furnace perlite expansion facility through mechanical redesign, controls tuning, and cross functional execution
- Analyzed and corrected critical bottlenecks with root cause analysis to eliminate recurring system stoppages
- Executed 4 to 5 factor DOEs in Minitab to tune furnace operating parameters and improve process stability
- Implemented density based feedback control to improve product consistency and reduce manual sampling
- Deployed a zero waste automated startup sequence, saving \$200K+ annually and scaling across Missouri and Virginia facilities
- Executed \$500K+ in CapEx, supporting equipment specification, vendor coordination, installation, and commissioning
- Partnered with operators and team leaders to validate process changes, troubleshoot equipment, and drive adoption of standard work

Tavis Corporation

Mariposa, CA

Project Engineer

Sep 2021 – May 2023

- Redesigned and validated a differential pressure transducer architecture across pressure ranges, cutting mechanical cost 50% and total system cost 30% on a \$40M+ aerospace program
- Developed a nonlinear diaphragm model where standard FEA was insufficient, validating predictions against test data
- Designed production ready components and assemblies in SolidWorks, including stress analysis and manufacturing drawings
- Developed and executed qualification test procedures for existing and prototype hardware to validate survival and operation under extreme thermal, vibration, shock, and overpressure conditions
- Led internal and customer facing design reviews, aligning requirements, tradeoffs, test results, and qualification deliverables
- Managed \$10M+ in hardware programs from concept through qualification and manufacturing integration, owning requirements, schedules, and technical deliverables
- Supported integrated hardware programs by answering customer questions, producing documentation, clarifying qualification data, and leading failure analysis
- Managed component obsolescence by assessing replacement options, qualification risk, design and cost impact, and documentation updates

Projects

Quantitative Research & Modeling System

- Built a modular machine learning system in Python for trading equities and options, spanning data ingestion, feature building, labeling, model training with LightGBM, backtesting, and live inference
- Engineered scalable Parquet pipelines to process large partitioned market datasets with clear schemas, staged artifacts, and reproducible outputs
- Developed option focused features from price, volume, Greeks, volatility, term structure, moneyness, and rolling returns
- Implemented a TOML based configuration system for repeatable experiments, module testing, and independent execution
- Built walk forward backtesting tools to evaluate model performance, strategy behavior, and execution assumptions